Covid Dataset Information

- Create a folder named covid_classification on Google Drive and open it. Right-click inside and create a new Google Colaboratory notebook.
- Go to the Colab notebook and rename it to image_classification.
- Create a folder named data in Google Drive and copy the covid_classification.zip file into it. We will unzip this file in the Colab notebook and save it inside this data folder.
- Inside the data folder, in the val, train, and test folders, there are X-ray images located in the viral_pneumonia, normal, and covid subfolders.
- In the Colab code:
    - First, select the T4 GPU.
    - Mount Google Drive.
    - Use the %cd .... command to specify the path to the working directory. To find this, check the folder structure on the left panel.
    - Install the ultralytics library.
    - Use an online image of a dog as the source, and apply YOLO's classification script to detect it – the model successfully identifies it.
    - Then, using Python code, we again classify the dog image. Due to repeated runs, we should check the latest prediction folder in the predict directory (like predict4).
    - To review the results, go to the corresponding folder in Google Drive, open the runs folder, and inspect the prediction results and graphs.
    - For the training part, unzip the dataset and automatically save it inside the data folder. Once this is done, there is no need to repeat it.
    - During model fitting, the operations are performed according to the number of epochs provided.
    - If the training is interrupted midway, there is a code block to resume from where it left off.
    - The training results can be found at: https://docs.ultralytics.com/modes/train. You can access and view the

results from both the left panel of Colab and the relevant folder in Google Drive.

- Create a folder named Inference in Google Drive and copy a few images from different folders in the previous dataset into it. These will be used for testing. Here, we use the best.pt file that contains the best model weights from the previous training.
- Provide the path of any image in the Inference folder and run a prediction on it. It will make a prediction based on the trained model.
- The results are provided at: https://docs.ultralytics.com/modes/predict – all prediction results can be reviewed here.
- With the last line of code, we specify the path of a single image and make a prediction.
